

Photon MCQS: Class-12 (Physics)

- The energy of a photon is E . If its wavelength is reduced to one-fourth of its original value, its energy becomes**
 - $E/4$
 - $E/2$
 - $2E$
 - $4E$
- Two photons have wavelengths λ and 2λ . The ratio of their momenta is**
 - 1:2
 - 2:1
 - 1:4
 - 4:1
- A photon has energy 6.2 eV. Its wavelength is approximately**
 - 100 nm
 - 200 nm
 - 300 nm
 - 400 nm
- A photon of frequency ν has momentum p . A photon of frequency 3ν has momentum**
 - $p/3$
 - p
 - $2p$
 - $3p$
- Which pair of quantities of a photon change when the photon passes from vacuum into a transparent medium?**
 - Frequency and energy
 - Frequency and momentum
 - Wavelength and momentum
 - Energy and frequency
- A photon has zero rest mass. Which statement is correct?**
 - It has zero momentum.
 - It cannot possess energy.
 - It possesses momentum despite having zero rest mass.
 - Its velocity must be zero.
- For a photon, $E = pc$. This relation follows because**
 - photon has rest mass
 - photon is charged
 - photon travels at c in vacuum
 - photon has zero energy
- Two monochromatic beams have equal power. Their photon frequencies are ν and 4ν . The ratio of the number of photons emitted per second is**
 - 1:4
 - 4:1
 - 1:2
 - 2:1

9. A beam contains N photons per second, each of energy E . If the frequency of every photon is doubled while the power remains constant, the number of photons per second becomes
- A. $2N$
 - B. N
 - C. $N/2$
 - D. $N/4$
10. Which photon has the greatest momentum?
- A. Radio-wave photon
 - B. Infrared photon
 - C. Visible-light photon
 - D. X-ray photon
11. The work function of a metal is 2.0 eV . Its threshold frequency is approximately
- A. $2.4 \times 10^{14} \text{ Hz}$
 - B. $4.8 \times 10^{14} \text{ Hz}$
 - C. $6.6 \times 10^{14} \text{ Hz}$
 - D. $9.6 \times 10^{14} \text{ Hz}$
12. A metal has threshold wavelength 500 nm . Its work function is approximately
- A. 1.24 eV
 - B. 2.48 eV
 - C. 3.10 eV
 - D. 4.96 eV
13. The threshold wavelength of a metal is λ_0 . If the work function is doubled, the new threshold wavelength becomes
- A. $2\lambda_0$
 - B. $\lambda_0/2$
 - C. $4\lambda_0$
 - D. unchanged
14. The threshold frequency of a metal is $5 \times 10^{14} \text{ Hz}$. Which radiation will not produce photoemission?
- A. $4 \times 10^{14} \text{ Hz}$
 - B. $5.5 \times 10^{14} \text{ Hz}$
 - C. $6 \times 10^{14} \text{ Hz}$
 - D. $8 \times 10^{14} \text{ Hz}$
15. A metal has work function 3 eV . Which photon can just produce photoemission?
- A. 2 eV
 - B. 2.5 eV
 - C. 3 eV
 - D. 1 eV
16. A metal has threshold wavelength 600 nm . Light of wavelength 900 nm is incident with extremely high intensity. What happens?
- A. Large photoelectric current is produced.
 - B. Electrons are emitted with low kinetic energy.
 - C. No photoelectric emission occurs.
 - D. Electrons are emitted after a time delay.
17. A metal has threshold frequency ν_0 . If incident frequency is $2\nu_0$, then the maximum kinetic energy is
- A. $h\nu_0$

- B. $2h\nu_0$
- C. zero
- D. $h\nu_0/2$

18. Einstein's photoelectric equation can be written as

- A. $K_{\max} = h\nu + \phi$
- B. $K_{\max} = h\nu - \phi$
- C. $K_{\max} = \phi - h\nu$
- D. $K_{\max} = h/\nu - \phi$

19. A photon of energy 5 eV strikes a metal whose work function is 2 eV. The maximum kinetic energy is

- A. 2 eV
- B. 3 eV
- C. 5 eV
- D. 7 eV

20. The work function of a metal is 2.2 eV. Light of photon energy 4.7 eV falls on it. The stopping potential is

- A. 2.5 V
- B. 4.7 V
- C. 6.9 V
- D. 2.2 V

21. Light of wavelength 400 nm falls on a metal of work function 2 eV. The stopping potential is approximately

- A. 0.10 V
- B. 1.10 V
- C. 2.10 V
- D. 3.10 V

22. The maximum kinetic energy of photoelectrons is 3.2×10^{-19} J. The stopping potential is approximately

- A. 1 V
- B. 2 V
- C. 3 V
- D. 4 V

23. A metal has work function ϕ . If the frequency of incident radiation is increased from ν to 2ν , the increase in maximum kinetic energy is

- A. $h\nu$
- B. $2h\nu$
- C. ϕ
- D. $h\nu - \phi$

24. A metal has work function 2 eV and incident photon energy 3 eV. The maximum speed of photoelectrons is proportional to

- A. $\sqrt{\frac{1\text{eV}}{m}}$
- B. $\frac{1\text{eV}}{m}$
- C. $\sqrt{\frac{2\text{eV}}{m}}$

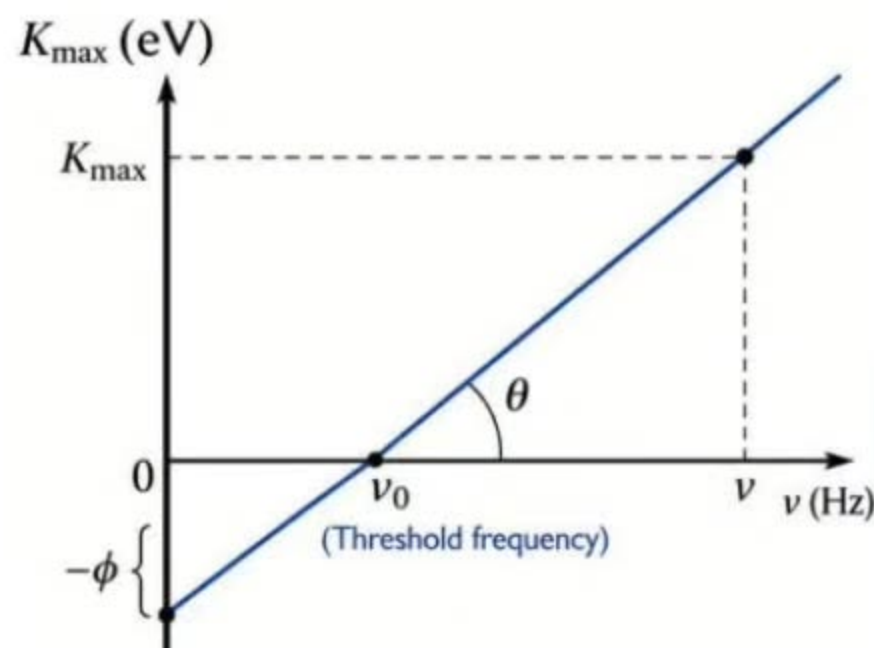
D. $\sqrt{\frac{3eV}{m}}$

25. For incident radiation above threshold frequency, increasing intensity while keeping frequency constant causes

- A. increase in $K_{\max}$
- B. decrease in $K_{\max}$
- C. increase in photoelectric current
- D. increase in work function

26. The intensity of incident light is remains unchanged. The stopping

- A. doubles
- B. becomes half
- C. remains unchanged
- D. becomes four times



doubled while frequency potential

27. The frequency of incident radiation is doubled while intensity is kept constant. Which statement is necessarily true?

- A. Photoelectric current doubles.
- B. Stopping potential doubles.
- C. Maximum kinetic energy increases by $h\nu$.
- D. Number of emitted electrons necessarily doubles.

28. For a fixed metal, two beams have frequencies ν and 2ν , both above thresholds. Which quantity is directly affected by the frequency?

- A. Work function
- B. Maximum kinetic energy
- C. Metal's threshold frequency
- D. Number of atoms in the metal

29. Which statement is incorrect?

- A. Increasing intensity increases photoelectric current.
- B. Increasing frequency increases maximum kinetic energy.
- C. Increasing intensity increases maximum kinetic energy.
- D. Below threshold frequency no photoemission occurs.

30. A graph of maximum kinetic energy $K_{\max}$ against frequency ν is a straight line. Which quantity is represented by its slope?

- A. h
- B. e
- C. h/e
- D. ϕ/e

31. For the same graph, the frequency-axis intercept represents

- A. work function
- B. stopping potential
- C. threshold frequency

D. Planck's constant

32. The K_{max} -versus- ν graph for two metals has the same slope but different frequency intercepts. This implies that the two metals have

- A. same work function
- B. different work functions
- C. same threshold frequency
- D. different Planck constants

33. The graph of stopping potential V_0 against frequency ν is

- A. $V_0 = \frac{h}{e} \nu - \frac{\phi}{e}$
- B. $V_0 = \frac{e}{h} \nu + \frac{\phi}{e}$
- C. $V_0 = h\nu + \phi$
- D. $V_0 = h\nu - \phi$

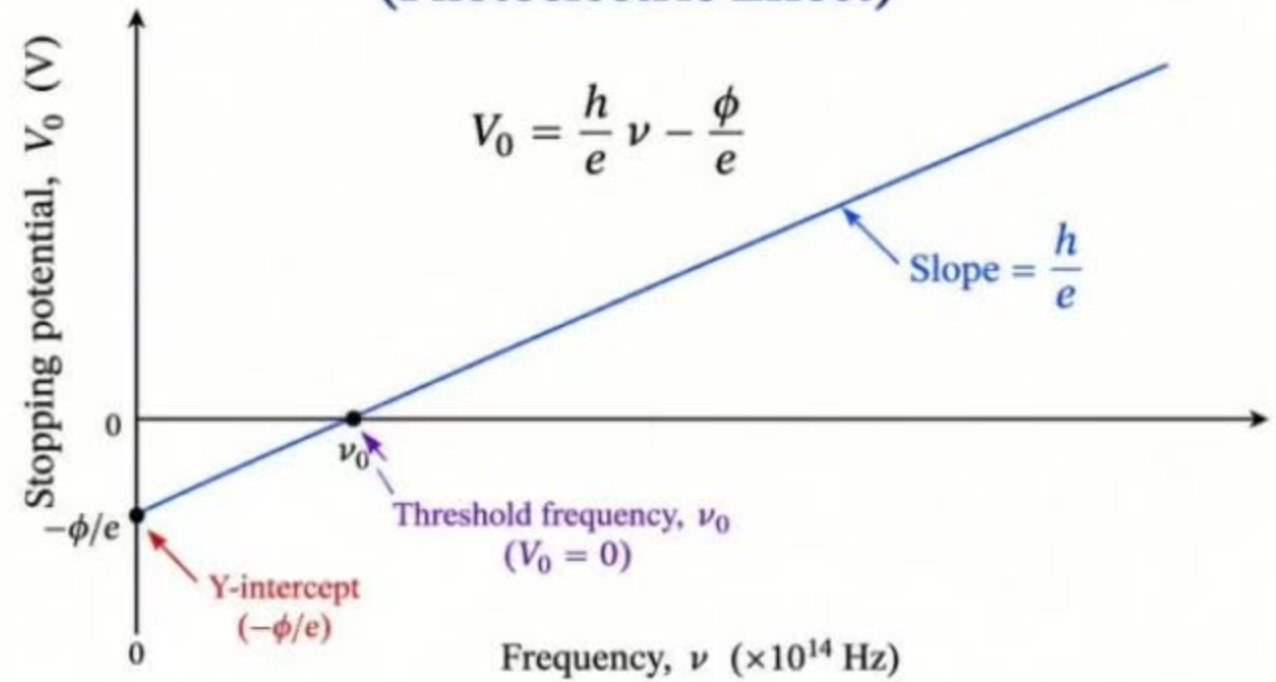
34. In a graph of V_0 versus ν , the slope is

- A. h
- B. $\frac{e}{h}$
- C. $\frac{h}{e}$
- D. $\frac{\phi}{e}$

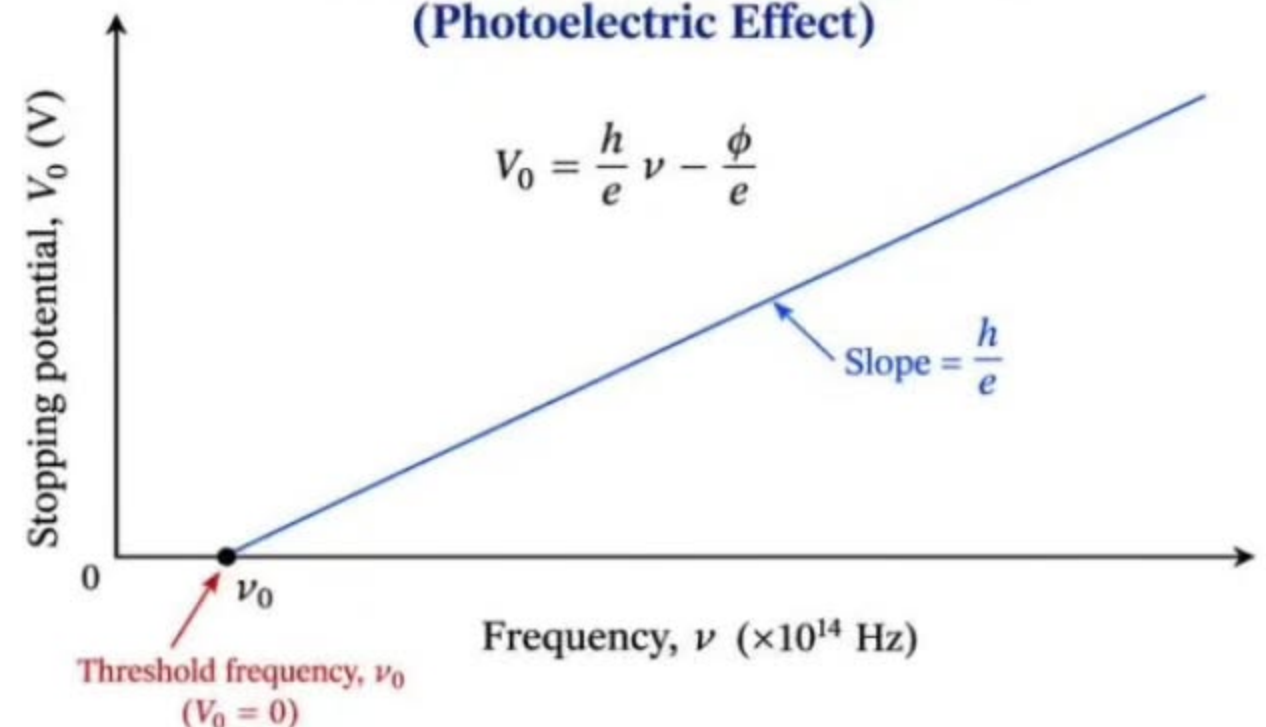
35. The y-intercept of a V_0 -versus- ν graph is

- A. ϕ
- B. $-\frac{\phi}{e}$
- C. $+\frac{\phi}{e}$
- D. $-\phi$

Stopping Potential V_0 vs Frequency (Photoelectric Effect)



Stopping Potential V_0 vs Frequency (Photoelectric Effect)



36. Data: ν (10^{14} Hz) = 5, 6, 7, 8; V_0 (V) = 0.20, 0.61, 1.02, 1.43. The approximate slope is

- A. $0.041 \text{ V}/(10^{14} \text{ Hz})$
- B. $0.41 \text{ V}/(10^{14} \text{ Hz})$
- C. $4.1 \text{ V}/(10^{14} \text{ Hz})$
- D. $41 \text{ V}/(10^{14} \text{ Hz})$

37. Three metals have V_0 -versus-frequency graphs that are parallel but have different frequency-axis intercepts. Which quantity is necessarily the same for all three?

- A. Work function
- B. Threshold frequency
- C. Planck's constant

D. Maximum kinetic energy

38. **For photoelectric emission, $V_0 = \frac{hc}{e} \left(\frac{1}{\lambda} \right) - \frac{\phi}{e}$. Therefore, a graph of V_0 versus $1/\lambda$ is**

- A. circular
- B. parabolic
- C. straight line
- D. exponential

39. **The slope of a V_0 versus $1/\lambda$ graph is**

- A. $\frac{h}{e}$
- B. $\frac{hc}{e}$
- C. $\frac{h}{c}$
- D. $\frac{e}{hc}$

40. **Three metals have threshold wavelengths $\lambda_{01} = 400$ nm, $\lambda_{02} = 500$ nm, $\lambda_{03} = 800$ nm. The ratio of their work functions is**

- A. 10:8:5
- B. 4:5:8
- C. 5:4:8
- D. 8:5:4

41. **Three metals have threshold wavelengths 400 nm, 500 nm, 800 nm. Which metal has the smallest work function?**

- A. Metal 1
- B. Metal 2
- C. Metal 3
- D. All are equal

42. **If the V_0 -versus- $1/\lambda$ lines for three metals are parallel, this means**

- A. all metals have the same work function
- B. all metals have the same threshold wavelength
- C. Planck's constant is the same for all
- D. the speed of light is different for each metal

43. **A photon of wavelength 500 nm falls on a metal having work function 1.5 eV. The maximum kinetic energy is approximately**

- A. 0.98 eV
- B. 1.50 eV
- C. 2.48 eV
- D. 3.98 eV

44. **A metal has threshold wavelength 620 nm. Light of wavelength 310 nm is incident on it. The maximum kinetic energy is**

- A. approximately equal to the work function
- B. approximately equal to twice the work function
- C. approximately equal to zero
- D. impossible to determine

45. **A metal has threshold frequency ν_0 . Incident radiation has frequency $5\nu_0$. The stopping potential is**

- A. ϕ/e

B. $4\phi/e$

C. $5\phi/e$

D. $\phi/5e$

46. A metal has stopping potential 1.5 V when illuminated by light of frequency 8×10^{14} Hz. The threshold frequency is approximately

A. 2.0×10^{14} Hz

B. 4.4×10^{14} Hz

C. 6.4×10^{14} Hz

D. 9.5×10^{14} Hz

47. A photon of energy E causes photoemission from a metal. If the work function is $E/3$, then the maximum kinetic energy is

A. $E/3$

B. $2E/3$

C. E

D. $3E$

48. The maximum kinetic energy of photoelectrons is 4 eV. If the work function is 2 eV, the energy of the incident photon is

A. 2 eV

B. 4 eV

C. 6 eV

D. 8 eV

49. A student argues: 'Increasing the intensity of light increases the energy of each photon.' The statement is

A. always true

B. always false

C. true only above threshold frequency

D. true only for UV radiation

50. Two beams have the same intensity but different frequencies, both above threshold. Which beam necessarily has

A. greater photon number if its frequency is greater

B. fewer photons per second if its frequency is greater

C. greater photoelectric current

D. greater work function

51. A very intense beam of frequency below threshold is incident on a metal. According to Einstein's photon theory

A. emission occurs after a long delay

B. emission occurs immediately

C. no photoemission occurs

D. electrons acquire half the photon energy

52. Suppose the frequency of incident light is exactly equal to the threshold frequency. Then the emitted photoelectrons have

A. maximum kinetic energy $h\nu_0$

B. maximum kinetic energy ϕ

C. zero maximum kinetic energy

D. infinite kinetic energy

53. If two metals have work functions $\phi_1 < \phi_2$, then their threshold frequencies satisfy
- $\nu_1 < \nu_2$
 - $\nu_1 > \nu_2$
 - $\nu_1 = \nu_2$
 - relation cannot be established
54. Which energy-conservation equation is correct for photoelectric emission?
- $h\nu = \phi + K_{\max}$
 - $h\nu = \phi - K_{\max}$
 - $h\nu = K_{\max} - \phi$
 - $h\nu = \phi/K_{\max}$
55. In Millikan's photoelectric experiment, a graph of stopping potential V_0 versus frequency ν is plotted. The value of Planck's constant is obtained from
- y-intercept
 - x-intercept
 - slope multiplied by e
 - slope divided by e
56. In an experiment, the graph of V_0 versus frequency has a slope smaller than expected. Which experimental result is most directly affected?
- Determination of Planck's constant
 - Charge of electron
 - Speed of light
 - Work function only
57. **Data: Frequency (10^{14} Hz) = 5,6,7,8; Stopping potential $V = 0.25, 0.66, 1.07, 1.48$. The threshold frequency is closest to**
- 3.0×10^{14} Hz
 - 4.4×10^{14} Hz
 - 5.0×10^{14} Hz
 - 6.0×10^{14} Hz
58. **The relativistic energy-momentum relation is $E^2 = p^2c^2 + m_0^2c^4$. For a photon, $m_0 = 0$. Therefore**
- $E = pc$
 - $E = mc^2$ only
 - $E = p/c$
 - $E = 0$
59. A photon has wavelength λ . Its equivalent momentum is
- $h\lambda$
 - h/λ
 - λ/h
 - hc/λ
60. A monochromatic beam has power P and frequency ν . The number of photons incident per second is
- $P/h\nu$
 - $P/(h\nu)$
 - $h\nu/P$
 - P/h
61. If the power of a monochromatic light source is P , its photon emission rate is N . If frequency is increased three times while power remains constant, the photon emission rate becomes

- A. $3N$
- B. N
- C. $N/3$
- D. $N/9$

62. A photon of frequency ν strikes an electron initially at rest in a metal. The electron emerges with maximum kinetic energy K . Which statement is correct?

- A. $K = h\nu + \phi$
- B. $K = h\nu - \phi$
- C. $K = \phi - h\nu$
- D. $K = h/\nu - \phi$

63. For a given metal, $K_{\max} = h\nu - \phi$. Differentiating with respect to frequency gives

$$\frac{dK_{\max}}{d\nu} =$$

- A. h
- B. ϕ
- C. $1/h$
- D. 0

64. The uncertainty in measuring the stopping potential is ΔV_0 . Since $K_{\max} = eV_0$, the corresponding uncertainty in maximum kinetic energy is

- A. $\Delta K = \Delta V_0/e$
- B. $\Delta K = e\Delta V_0$
- C. $\Delta K = e/\Delta V_0$
- D. $\Delta K = 0$

65. If the stopping potential is plotted against $1/\lambda$, then

$$\frac{dV_0}{d\left(\frac{1}{\lambda}\right)}$$

is equal to

- A. h/e
- B. hc/e
- C. e/hc
- D. hc

66. A metal has threshold wavelength λ_0 . For incident wavelength $\lambda < \lambda_0$, the stopping potential is

- A. $\frac{hc}{e} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$
- B. $\frac{hc}{e} \left(\frac{1}{\lambda_0} - \frac{1}{\lambda} \right)$
- C. $\frac{h}{e} (\lambda - \lambda_0)$
- D. $\left(\frac{hc}{e} \right) \lambda \lambda_0$

67. For two metals A and B, $\phi_A = 2\phi_B$. Their threshold frequencies satisfy

- A. $\nu_A = 2\nu_B$
- B. $\nu_A = \frac{\nu_B}{2}$
- C. $\nu_A = 4\nu_B$
- D. $\nu_A = \nu_B$

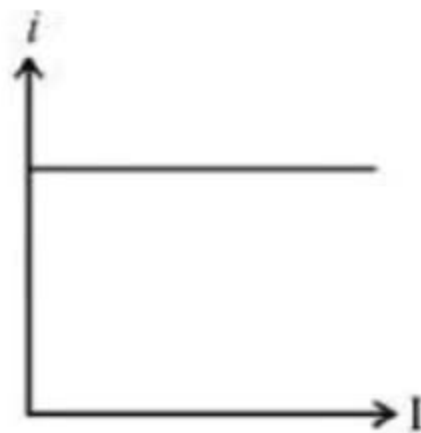
68. Two monochromatic beams of equal intensity have wavelengths λ and 2λ . Which beam contains more photons per second?
- λ -beam
 - 2λ -beam
 - Both contain equal photons
 - Cannot be determined
69. A metal has threshold wavelength 600 nm. Two lights of wavelengths 300 nm and 450 nm illuminate it. The ratio of maximum kinetic energies is
- 1:1
 - 1:2
 - 3:1
 - 5:2
70. For the same metal, light of frequency $2\nu_0$ produces stopping potential V . Light of frequency $3\nu_0$ produces stopping potential
- $V/2$
 - V
 - $3V/2$
 - $2V$
71. A metal has threshold frequency ν_0 . For incident frequency $4\nu_0$, the stopping potential is V . If the frequency is changed to $7\nu_0$, the stopping potential becomes
- $7V/4$
 - $2V$
 - $3V/2$
 - $V + 3h\nu_0/e$
72. A metal is illuminated successively by wavelengths 400 nm, 500 nm, and 600 nm. Photoemission occurs for the first two but not the third. Its threshold wavelength must satisfy
- $\lambda_0 < 400 \text{ nm}$
 - $400 < \lambda_0 < 500 \text{ nm}$
 - $500 < \lambda_0 < 600 \text{ nm}$
 - $\lambda_0 > 600 \text{ nm}$
73. A photoelectric experiment gives $V_0 = 0$ for $\nu < 5 \times 10^{14} \text{ Hz}$, and $V_0 > 0$ for $\nu > 5 \times 10^{14} \text{ Hz}$. The threshold frequency is
- 0
 - $5 \times 10^{14} \text{ Hz}$
 - greater than $5 \times 10^{14} \text{ Hz}$
 - impossible to determine
74. Two metals A and B have threshold frequencies $4 \times 10^{14} \text{ Hz}$ and $6 \times 10^{14} \text{ Hz}$ respectively. For incident frequency $8 \times 10^{14} \text{ Hz}$, $K_A:K_B$ is
- 1:1
 - 2:1
 - 1:2
 - 4:3
75. A photon of energy 10 eV strikes a metal with work function 4 eV. If the emitted electron subsequently passes through a potential difference of 2 V in the accelerating direction, its kinetic energy becomes

- A. 4 eV
- B. 6 eV
- C. 8 eV
- D. 12 eV

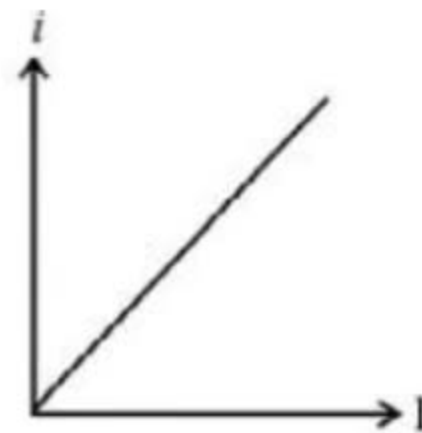
76. If the intensity of incident radiation is reduced to one-fourth while frequency remains above threshold, the maximum kinetic energy

- A. becomes one-fourth
- B. becomes four times
- C. remains unchanged
- D. becomes half

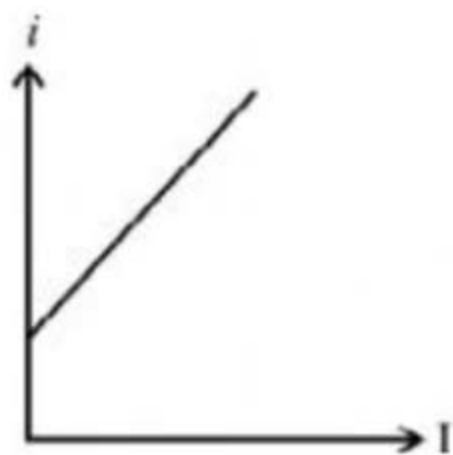
77. Which of the following graphs represents correctly the variation of photoelectric current (i) with intensity (I) of incident radiations:



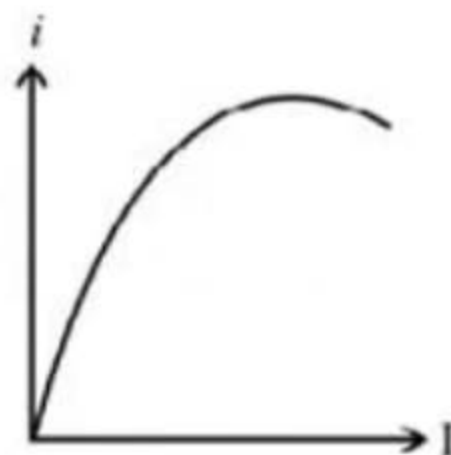
A.



B.

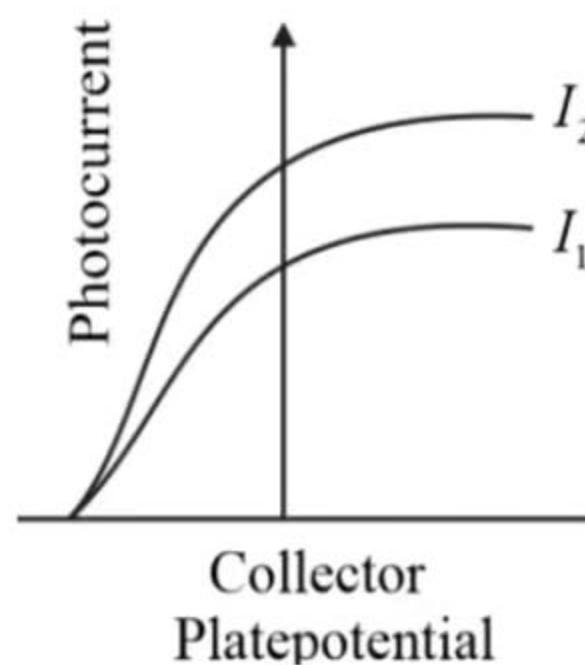


C.



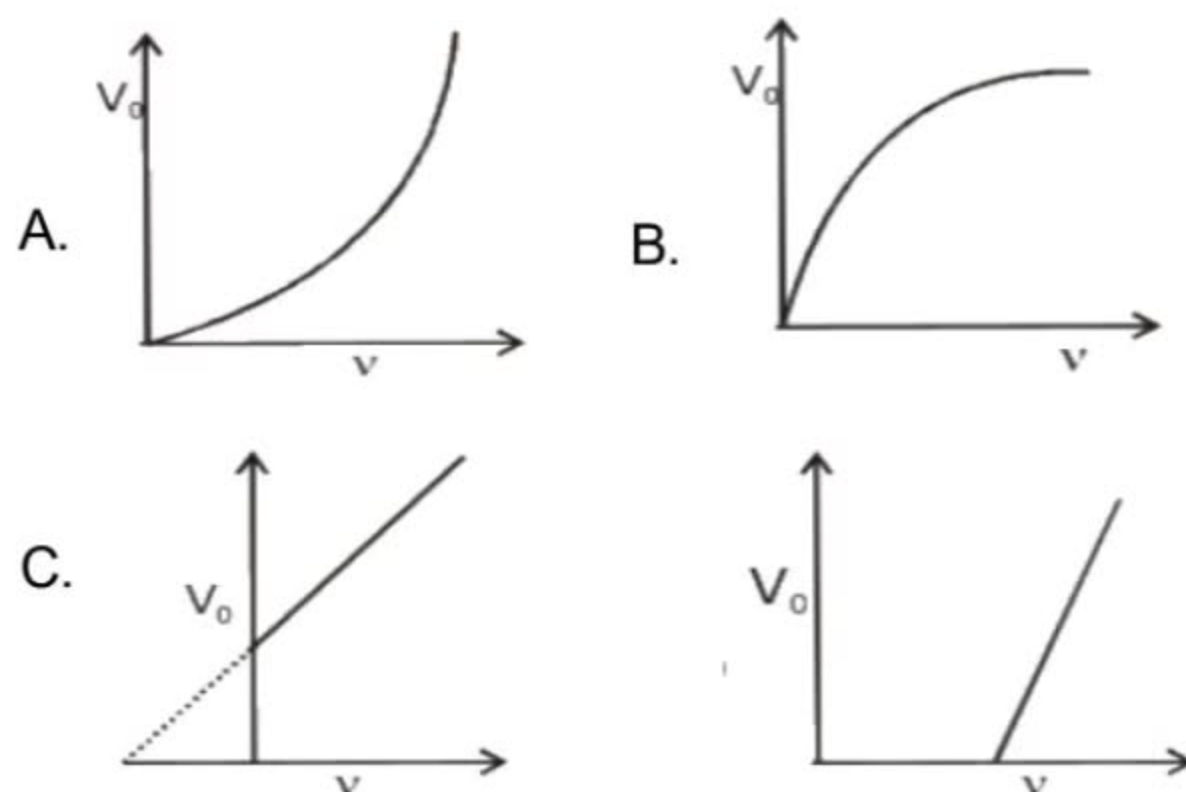
D.

78. From the following graph of photo current against collector plate potential, for two different intensities of light I_1 and I_2 , one can conclude:

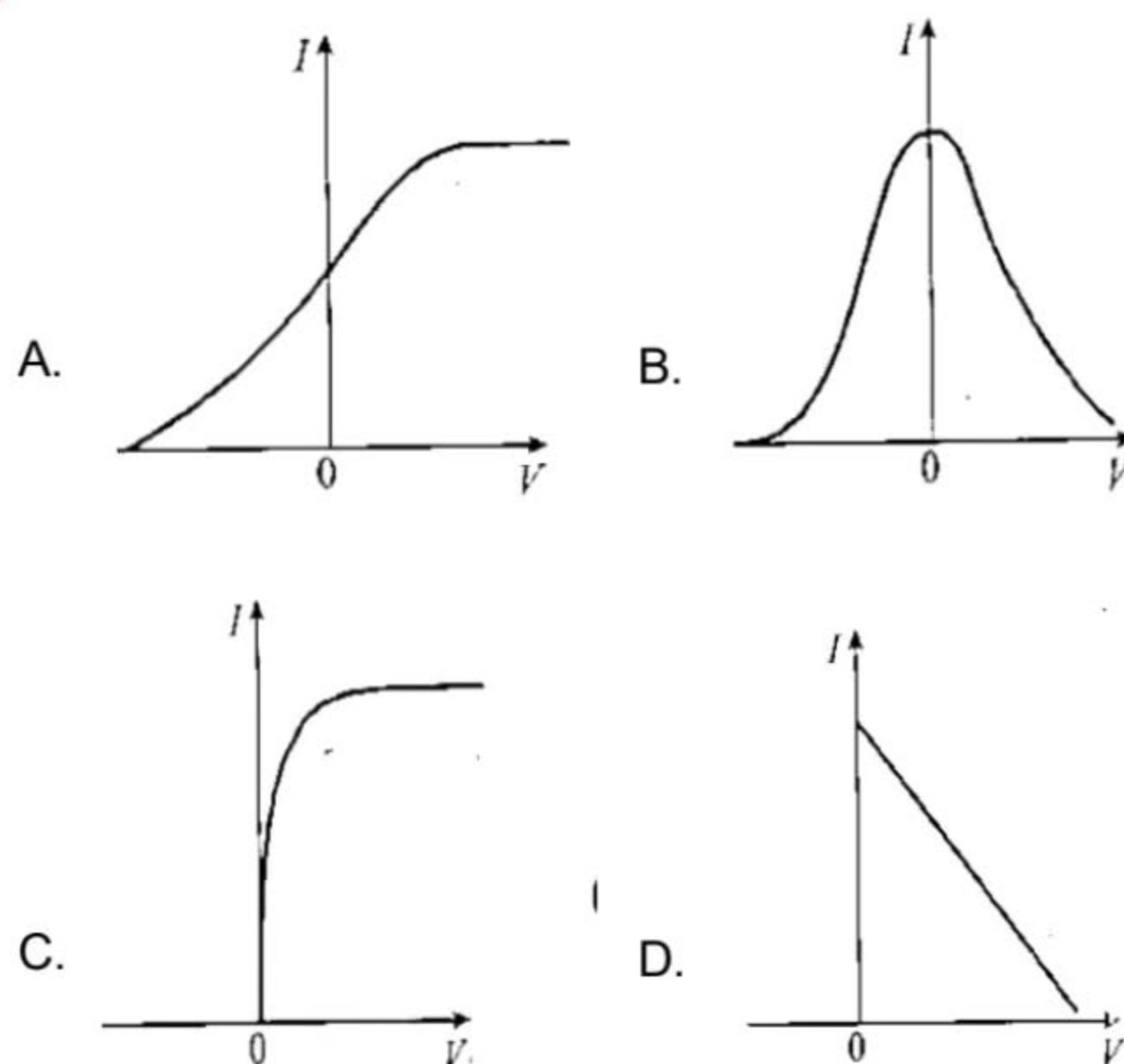


- A. $I_1 = I_2$
- B. $I_1 > I_2$
- C. $I_1 < I_2$
- D. comparison is not possible.

79. For a photographic cell, the graph showing the variation of cut off voltage or maximum kinetic energy of ejected photo – electrons with frequency of incident light is:



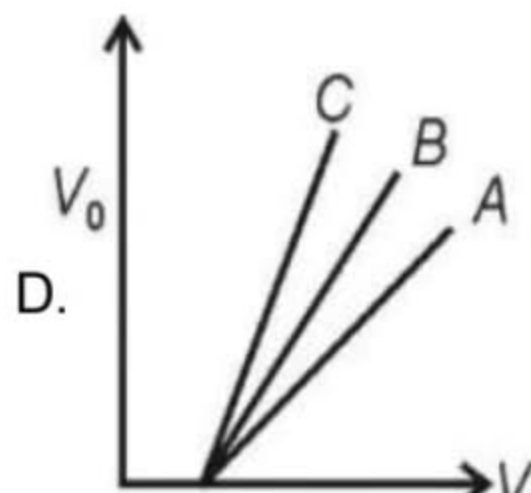
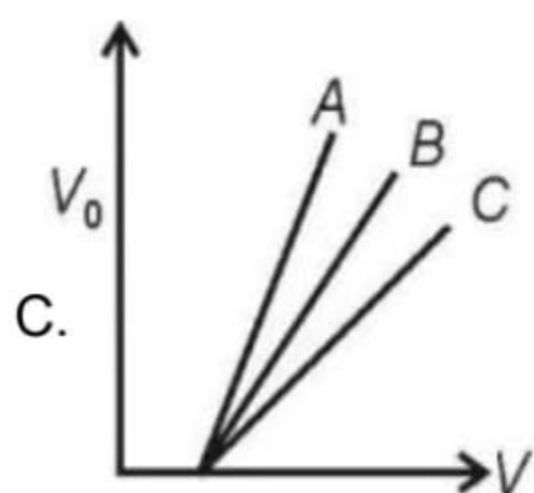
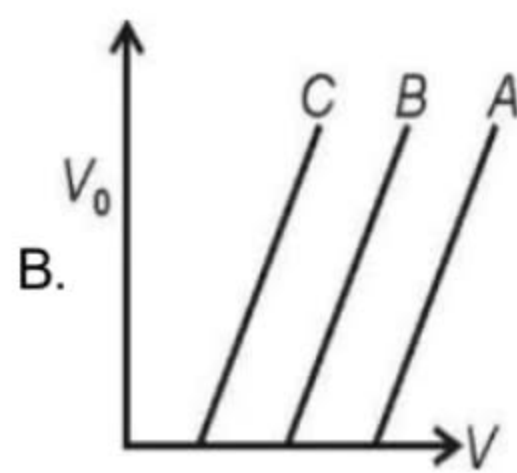
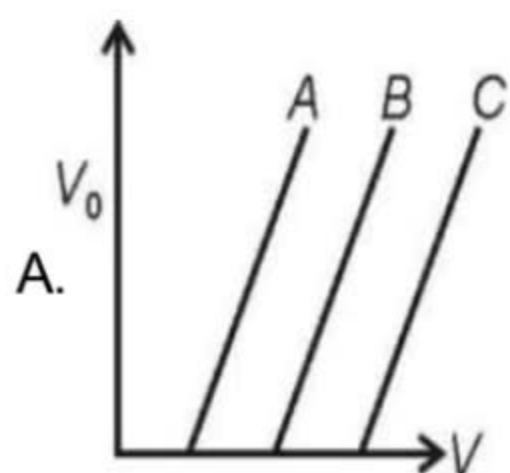
80. Which of the following graphs gives the variation of photoelectric current I_p with voltage (V) applied to the electrodes of a photocell



81. Which observation provides the strongest evidence that electromagnetic radiation transfers energy in discrete packets?

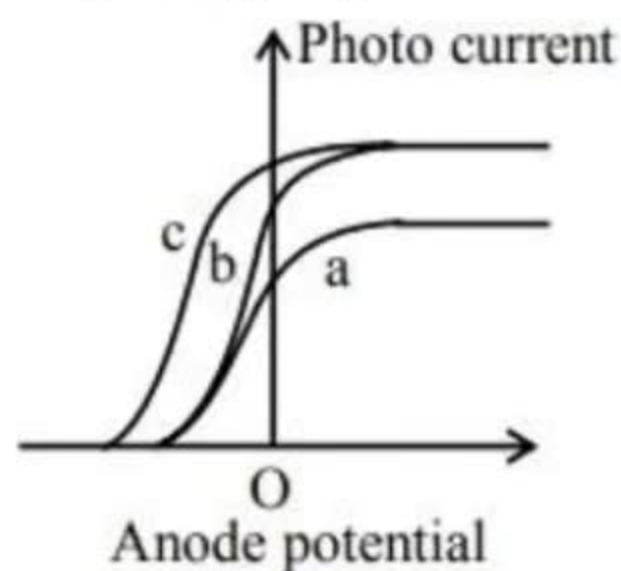
- A. Diffraction
- B. Interference
- C. Photoelectric effect
- D. Polarization

82. The work function of three different metals A, B and C are ϕ_A , ϕ_B and ϕ_C where $\phi_A > \phi_B > \phi_C$. The graph between stopping potential V_0 and frequency f of the incident radiation from them would look like:

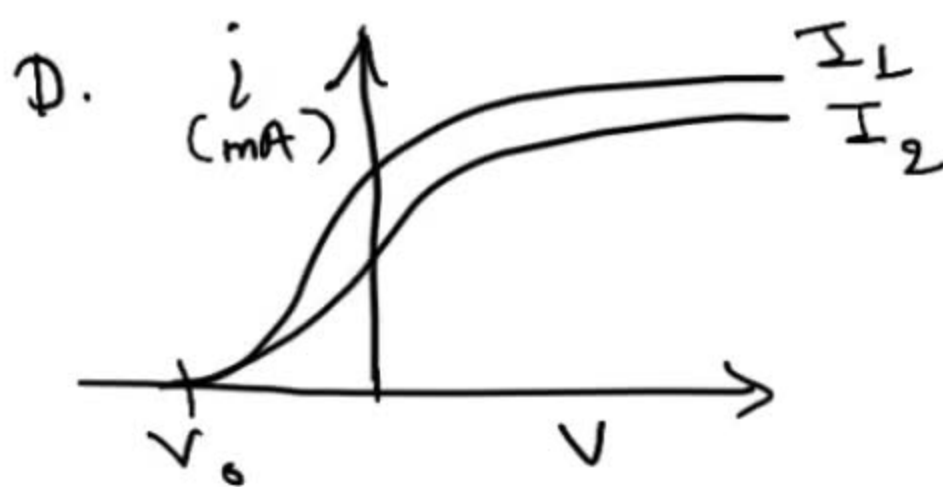
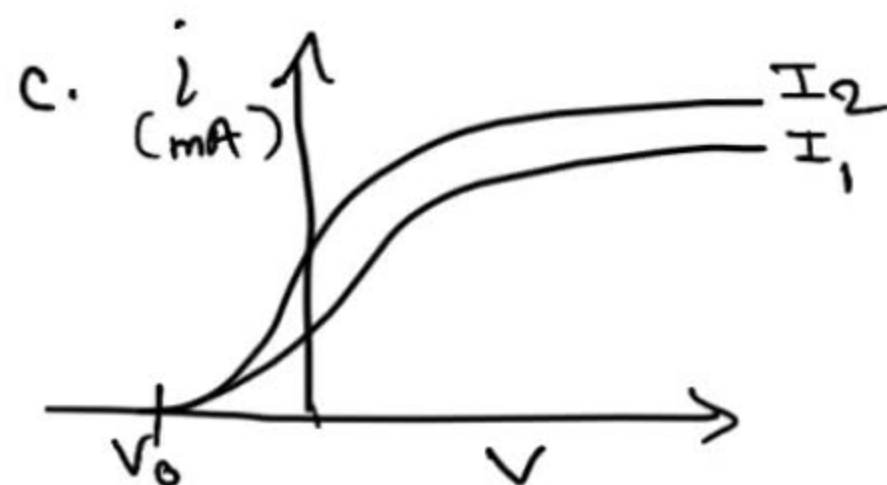
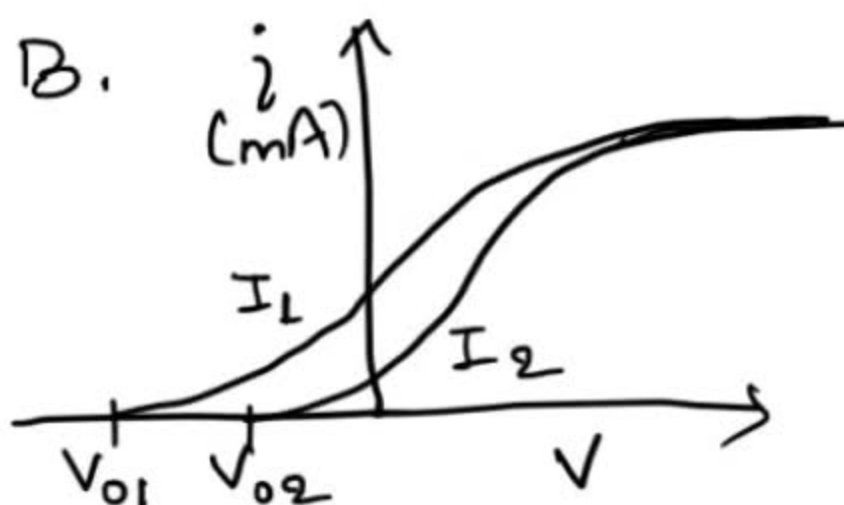
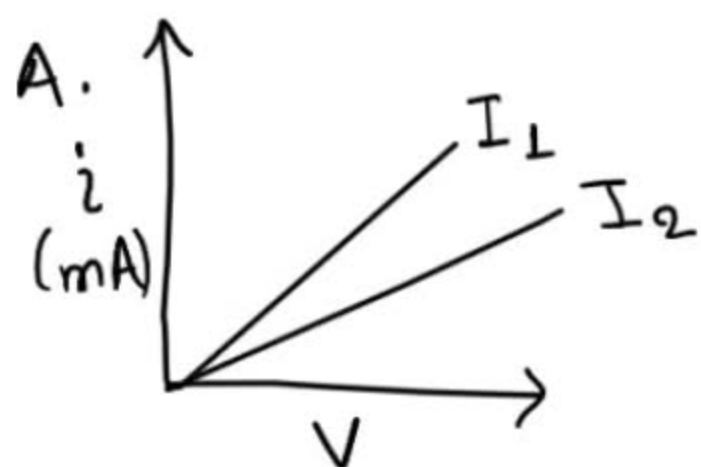


83. The figure shows the variation of photocurrent with anode potential for a metal surface for three different radiations. Let I_a , I_b and I_c be the intensities and f_a , f_b and f_c be the frequencies for the curves respectively. Then the true statement is

- A. $f_a = f_b$ and $I_a = I_b$ B. $f_a = f_c$ and $I_a = I_c$
 C. $f_a = f_b$ and $I_a \neq I_b$ D. $f_b = f_c$ and $I_b = I_c$



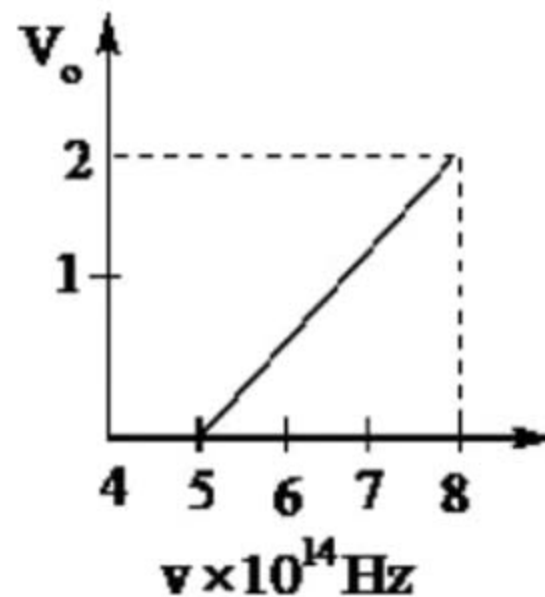
84. Variation of photoelectric current with collector plate potential for different intensities ($I_1 > I_2$) at a fixed frequency is



85. If the frequency of light in photoelectric experiment is doubled, the stopping potential will

- A. be halved
- B. become less than double
- C. be doubled
- D. become more than double

86. The stopping potential (V_0) versus frequency plot of a substance is shown in fig. The threshold wavelength is



- A. 4000 Å
- B. 6000 Å
- C. 5000 Å
- D. cannot be estimated from given data

87. Light of two different frequencies whose photons have energies 1 eV and 2.5 eV respectively illuminate a metallic surface whose work function is 0.5 eV successively. Ratio of maximum speed of emitted electrons will be

- A. 1:4
- B. 1:2
- C. 1:1
- D. 1:5

88. Stopping potential for photoelectron is 1 V. Maximum KE

- A. 1 eV
- B. 2 eV
- C. 0.5 eV
- D. 5 eV

89. When light falls on a photosensitive surface electron are emitted from the surface. The kinetic energy of these electrons does not depend on the

- A. wavelength of light
- B. Frequency of light
- C. Type of material used for the surface
- D. intensity of light

90. The work function of Cesium (Cs), Potassium (K) and Sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV, which of this photosensitive surface may emit photoelectrons?

- A. K only
- B. Na only
- C. Cs only
- D. Both Na and K

Useful constants:

- $h = 6.626 \times 10^{-34} \text{ Js}$
- $c = 3 \times 10^8 \text{ m/s}$
- $e = 1.6 \times 10^{-19} \text{ C}$

Quickly recall:

$$E = hf \Rightarrow E \propto f \Rightarrow \frac{E_1}{E_2} = \frac{f_1}{f_2}$$

$$E = hf = \frac{hc}{\lambda} \Rightarrow E \propto \frac{1}{\lambda} \Rightarrow \frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1}$$

$$p = \frac{h}{\lambda} \Rightarrow p \propto \frac{1}{\lambda} \Rightarrow \frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1}$$

$$p = \frac{h}{\lambda} = \frac{hf}{c} \Rightarrow p \propto f \Rightarrow \frac{p_1}{p_2} = \frac{f_1}{f_2}$$

Power of monochromatic beam, $P = nhf$ where $n = \text{no. of photons emitted per sec.}$

$$\Rightarrow n = \frac{P}{hf} \Rightarrow n \propto \frac{1}{f} \Rightarrow \frac{n_1}{n_2} = \frac{f_2}{f_1}$$

$$\Rightarrow n \propto \lambda \Rightarrow \frac{n_1}{n_2} = \frac{\lambda_1}{\lambda_2}$$

$$E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$$

$$\phi(\text{eV}) = \frac{f_0}{2.4 \times 10^{14}} \Rightarrow f_0 = \phi(\text{eV}) \times 2.4 \times 10^{14}$$

$$\lambda_0(\text{nm}) = \frac{1240}{\phi(\text{eV})}$$

- Photon energy: $E = h\nu = hc/\lambda$
- Photon momentum: $p = h/\lambda = E/c$
- Work function: $\phi = h\nu_0 = hc/\lambda_0$
- Einstein photoelectric equation: $K_{\text{max}} = h\nu - \phi$
- Stopping potential: $K_{\text{max}} = eV_s$
- $V_s = (h/e)\nu - \phi/e$
- $V_s = (hc/e)(1/\lambda) - \phi/e$
- Slope of V_s versus $\nu = h/e$
- Slope of V_s versus $1/\lambda = hc/e$
- For K_{max} versus ν , slope = h and frequency intercept = ν_0
- Intensity mainly affects photoelectric current; frequency determines photon energy and K_{max} (for fixed metal).
- Uncertainty relation used in Q67: $\Delta K_{\text{max}} = e \Delta V_s$.